

The Power of Visualization: Using NeXTA4GMNS for Trajectory Display

Cafer Avci^{1¶}, Yajun Liu², and Xuesong (Simon) Zhou²

¹ School of Civil and Environmental Engineering, Cornell University, Ithaca, NY, 14853, USA ² School of Sustainable Engineering and the Built Environment, Arizona State University, Tempe, AZ, 85281, USA ¶ Corresponding author

DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

Software

- [Review](#) ↗
- [Repository](#) ↗
- [Archive](#) ↗

Editor: ↗

Submitted: 03 March 2026

Published: unpublished

License

Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License ([CC BY 4.0](https://creativecommons.org/licenses/by/4.0/)).

Summary

Contemporary transportation modeling requires accurate and intuitive visualization tools to effectively interpret increasingly complex and high-dimensional data, including multi-resolution network topologies, time-varying traffic states, detailed vehicle trajectories, and large-scale simulation outputs. We present NeXTA4GMNS (NeXTA), an open-source platform designed specifically for network and trajectory visualization that fully embraces the General Modeling Network Specification (GMNS) standard ([GMNS Development Team, 2020](#)). NeXTA provides editable, multi-resolution node-link representations of transportation networks, enabling the analysis of link performance—such as travel time, traffic volume, and speed—across different time horizons. In addition to link-level metrics, it supports the visualization of vehicle trajectories, tensor-based data structures, and shortest path computations. Unlike general-purpose GIS tools, NeXTA is specifically designed for scientific transportation modeling, offering advanced capabilities for trajectory analysis, origin-destination trip and path exploration, and seamless integration with simulation frameworks such as FHWA's AMS and dynamic traffic assignment (DTA) models ([Yelchuru et al., 2017](#)).

Statement of Need

Transportation systems have become increasingly complex in recent decades, making traditional methods for analyzing traffic flow—such as manual counts, loop detectors, and periodic surveys—insufficient for capturing the dynamic behavior of modern travel patterns ([Papageorgiou, 1991](#)). These methods often lack the spatial and temporal resolution needed to support detailed modeling, validation, and policy analysis in contemporary traffic systems.

Recognizing this limitation, GMNS community has highlighted the need for standardized network representations, along with visualization platforms that can support the scale and complexity of real-world datasets ([GMNS Development Team, 2020](#)). Visualization tools that facilitate the integration of network topology, traffic dynamics, and simulation outputs are especially critical.

NeXTA was developed in response to these needs. It is a GMNS-compliant, open-source visualization platform designed for transportation network analysis and trajectory data interpretation. Rather than focusing solely on map rendering, NeXTA supports multi-resolution node-link representations and incorporates tensor-based data structures to facilitate the analysis of spatiotemporal traffic patterns grounded in established traffic flow theory ([Lighthill & Whitham, 1955](#); [Richards, 1956](#)).

Feature	NeXTA	QGIS
Network Representation	GMNS-compliant, node-link model	GIS shapefiles and layers
Trajectory Analysis	Built-in support for microscopic trajectory data (e.g., NGSIM)	Requires external plugins and preprocessing
Traffic Simulation Integration	Native support for simulation outputs (e.g., DTALite (Zhou et al., 2012), SUMO (with post-processing) (Behrisch et al., 2011))	Limited support; not transportation-specific
Network Editing	Editable, hierarchical, and multi-resolution network views	Basic vector layer editing, not traffic-centric
Tensor-Based Modeling	Supports tensor structures for trajectory processing	Not available
Usability for Transportation Practitioners	Tailored for planners, researchers, and educators in traffic modeling	General-purpose GIS users

Its utility lies in enabling users to explore traffic dynamics, examine vehicle trajectories, and modify network structures for simulation input or scenario testing. Compared to general-purpose GIS tools, NeXTA offers built-in support for transportation-specific tasks such as interpreting outputs from microscopic simulation models and editing hierarchical transportation networks. Datasets such as the Next Generation Simulation (NGSIM) (Federal Highway Administration, 2007) provide detailed vehicle trajectories that can be difficult to analyze without specialized tools. NeXTA helps bridge this gap by enabling interactive visualization and analysis of such high-resolution data, supporting both research and practice in transportation modeling and simulation.

Software Description

Core Features and Capabilities

Multi-resolution network visualization and editing

One of NeXTA's most powerful features is its support for seamless transitions between different levels of detail. The framework handles everything from detailed lane-level analysis to broad corridor-level planning perspectives. Users can edit and analyze networks at the link level for microsimulation work, aggregate up to path-level analysis for route choice studies, or zoom out to zonal-level analysis for strategic planning—all within the same interface.

Tensor-based representation of trajectories

Rather than treating vehicle trajectories as simple sequences of points, NeXTA represents them as multi-dimensional tensors that capture the full richness of movement data. This formulation enables more compact data storage, efficient data manipulation, and supports advanced analytical operations compared to traditional trajectory representations. It allows us to have more efficient storage, faster manipulation, and more sophisticated analysis capabilities than traditional approaches.

Cross-resolution and multi-view trajectory visualization

NeXTA excels at showing individual vehicle trajectories unfold in real-time while simultaneously observing aggregate flow patterns across the entire network. This capability is crucial for understanding how microscopic driving behaviors aggregate into macroscopic traffic phenomena—a fundamental challenge in traffic flow theory (Gazis, 2002; Treiber & Kesting, 2013).

Integration with simulation frameworks

The platform seamlessly integrates with popular traffic simulation tools, supporting data exchange and synchronization. This integration philosophy aligns with FHWA's emphasis on effective integration of analysis, modeling, and simulation tools in transportation planning (Nevers et al., 2013). This makes it particularly valuable for developing and testing advanced transportation network management strategies, where users need to quickly visualize the impacts of different control policies or infrastructure modifications (Papageorgiou et al., 2003).

Multi-dimensional Matrix Representations

At the heart of NeXTA's analytical power is its sophisticated mathematical representation of traffic data. We capture agent movements using a trajectory tensor $X(a)$ that includes comprehensive information about each vehicle or traveler:

- Unique agent identity and classification
- Origin-destination pair information
- Detailed route specifications and path choices
- Precise departure and arrival timing
- Vehicle characteristics and driver behavior parameters

To support robust optimization and calibration procedures, we also maintain a baseline prior estimate $X_h(a)$ that provides regularization when working with noisy or incomplete data—a common challenge when dealing with real-world trajectory datasets. Complementing this individual-level data, measurement tensor $Y(i, j, t, m)$ tracks the real-time state of network links, capturing flow rates, traffic density, average speeds, and other performance metrics. The connection between individual trajectories and aggregate network conditions is established through proportional mapping operators that have their roots in classical traffic assignment theory (Beckmann et al., 1956; Wardrop & Whitehead, 1952).

Computational Graph for Traffic Modeling

The mathematical relationships underlying NeXTA draw from decades of traffic flow theory while incorporating modern computational approaches. One of NeXTA's most innovative features is its use of computational graphs to structure traffic modeling calculations, as illustrated in Figure 1. This approach enables efficient forward and backward propagation of gradients, supporting optimization-based calibration of travel times and flows.

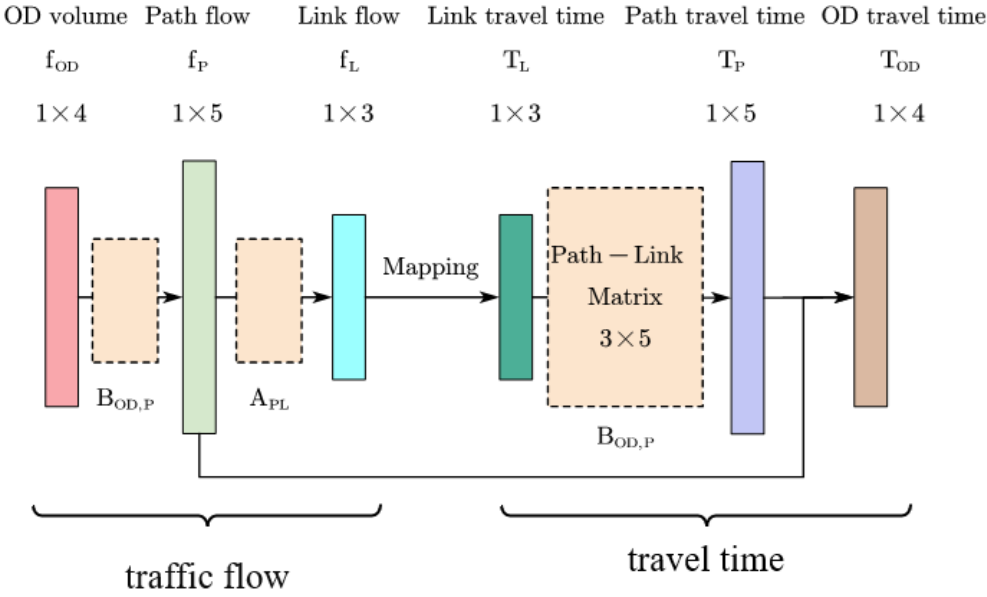


Figure 1: Layered computational graph representing OD-to-link relationships through path-based assignments and travel time transformations.

99 The multi-layer structure transforms data systematically between different network elements.
100 In **Layer 1**, we convert origin-destination flows F_{OD} into path flows f_P using an OD-to-Path
101 assignment matrix B : $f_P = B \cdot F_{OD}$. This step captures how travelers choose among
102 alternative routes, incorporating principles from discrete choice theory (Ben-Akiva & Lerman,
103 1985; McFadden, 1974).

104 **Layer 2** then aggregates these path flows to produce link flows f_L using a Path-to-Link
105 incidence matrix A : $f_L = A \cdot f_P$. This is where individual route choices combine to create the
106 traffic loads we observe on network links.

107 The reverse transformation happens in **Layer 3**, where link travel times T_L are aggregated into
108 path travel times T_P using the transpose relationship: $T_P = A^T \cdot T_L$. Finally, **Layer 4** maps
109 these path travel times back to OD-level travel times using $T_{OD} = B \cdot T_P$.

110 This computational graph approach isn't just mathematically elegant—it's also highly practical.
111 The structure naturally supports gradient-based optimization methods, making it much easier
112 to calibrate complex traffic models against observed data.

113 The beauty of this layered approach becomes clear when you consider how it bridges abstract
114 mathematical modeling with practical, user-friendly analysis tools. Every component of the
115 computational graph shown in Figure 1 maps directly to specific visualization features
116 within NeXTA's interface, creating a seamless connection between theory and practice.

117 Table ?? shows exactly how this mapping works. Elements like OD flows, path flows, and
118 travel times aren't just mathematical abstractions—they become interactive visual components
119 that help users understand their transportation systems. This connection between rigorous
120 modeling and intuitive visualization is what makes NeXTA particularly effective for model
121 calibration, validation, and communication with stakeholders who may not have deep technical
122 backgrounds.

Table 2: Modeling Elements and Visualization Support in NeXTA {#tab:modeling_NeXTA}

Modeling Element	NeXTA Visualization Feature
OD Flows (F_{OD})	Interactive OD pair filtering and summary statistics
Path Flows (f_P)	Comprehensive route/path analysis view
Link Flows (f_L)	Real-time link performance view (flow, speed, density)
Link Travel Times (T_L)	Dynamic heatmaps and link-level metrics dashboard
Path Travel Times (T_P)	Aggregated trajectory visualizations with time profiles
OD Travel Times (T_{OD})	OD pair-level average travel time matrices
Tensor Representation ($X(a)$)	Multi-dimensional filtering and agent tracking

123 **Practical Applications and Examples**

124 **OD-Based Trajectory Analysis**

125 NeXTA offers a scientifically grounded environment for exploring transportation dynamics
126 across spatial and temporal dimensions. At the core of this capability is the OD pair and
127 path-based vehicle trajectory filtering interface, illustrated in Figure [Figure 2](#). This tool allows
128 users to dissect complex movement patterns by filtering trajectories based on origin-destination
129 zones, departure time windows, and vehicle classifications—enabling focused analysis of freight
130 activity, peak-hour congestion, or regional travel demand.

131 These filtering capabilities are tightly coupled with the platform's multi-layer analytical
132 framework. Filtering by OD zones corresponds to **Layer 1**, where flows are interpreted
133 in terms of traveler route choices across alternative paths. Once trajectories are narrowed,
134 users can transition to **Layer 2** analyses, examining how those path-based movements translate
135 into link-level congestion and flow conditions. In this view, NeXTA allows users to visualize
136 how route-level decisions—aggregated from large-scale vehicle trajectories—influence link
137 performance metrics such as volume, speed, and travel time.

138 The tool also supports downstream analyses aligned with **Layers 3 and 4**, enabling users
139 to assess how link-level conditions affect cumulative travel experiences across entire paths
140 and back to OD-level performance summaries. This layered perspective ensures that filtered
141 trajectories are not only visualized, but also quantitatively analyzed within a consistent system
142 that spans OD flows, path assignments, and link-level dynamics.

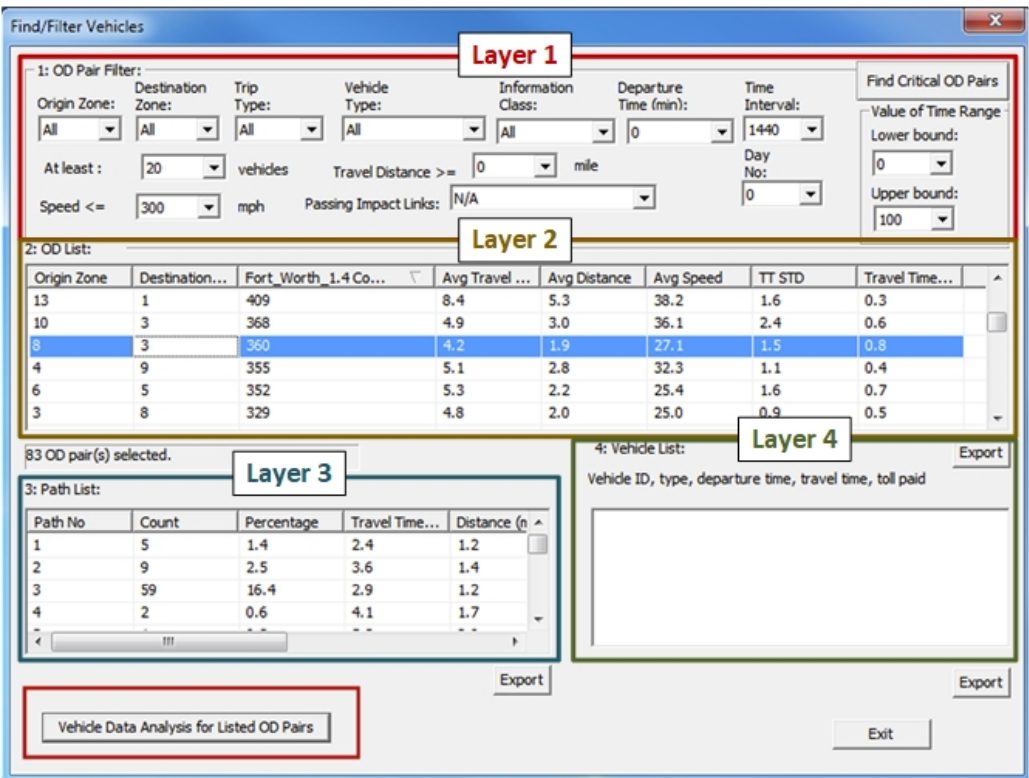


Figure 2: OD Pair and Path-based vehicle trajectory filtering in NeXTA, showing the interface for analyzing travel patterns across different origin-destination pairs.

Building on this structured framework, NeXTA's approach to route-based trajectory visualization aligns closely with established practices in the field—particularly those outlined in the FHWA's comprehensive *Trajectory Processor User's Guide* (Federal Highway Administration, 2011). At the same time, NeXTA advances these methods by offering a standardized GMNS-compliant environment that integrates OD-based filtering with path-level analysis and link-level performance metrics.

High-Resolution Empirical Data Analysis

Figure 3 showcases NeXTA's capabilities when working with high-resolution empirical datasets. In this example, we're visualizing trajectories with the NGSIM I-101 dataset, focusing on Lane 2 with speed and spacing filters applied to extract vehicle behavior patterns within specific spatial and temporal constraints.

This type of detailed analysis opens up entirely new possibilities for understanding traffic flow phenomena. Users can study lane utilization patterns to optimize lane management strategies, identify stop-and-go wave propagation to understand congestion formation, and analyze acceleration profiles to calibrate car-following models. The visual rendering incorporates dynamic overlays—speed profiles, trajectory segment colors, density indicators—that make it much easier to spot important phenomena like shockwave formation, merging conflicts, or capacity bottlenecks.

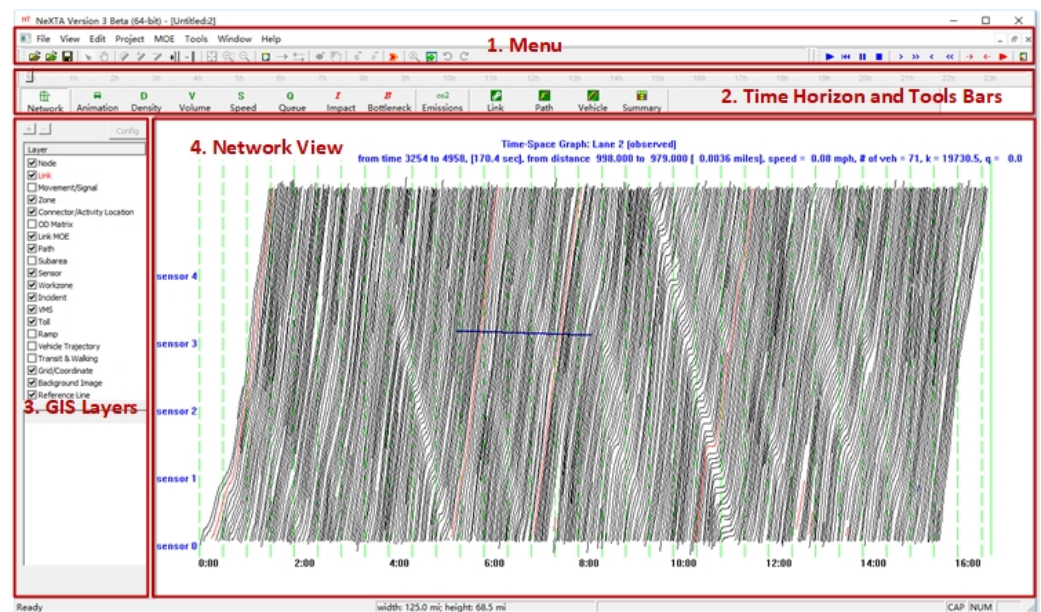


Figure 3: 2D NGSIM 101 Trajectory Visualization of Lane 2 with distance and speed selection filters applied to highlight specific vehicle behavior patterns.

This capability proves particularly valuable when you're trying to validate microsimulation results against observed data or calibrate model parameters based on real-world traffic dynamics. Traditional approaches to this kind of validation often rely on aggregate statistics that can mask important behavioral details. NeXTA's trajectory-level visualization allows for much more nuanced comparisons between observed and simulated behavior, making it an essential bridge between theoretical model development and practical deployment.

The integration with empirical datasets like NGSIM also supports more advanced applications, including the development of machine learning models for traffic prediction (Kim et al., 2020), the calibration of emerging connected and automated vehicle models (Shladover et al., 2012), and the validation of traffic flow theories under real-world conditions.

Demonstration and Learning Resources

To help users get the most out of NeXTA's capabilities, we've developed comprehensive demonstration materials. A detailed walkthrough video is available on YouTube: https://www.youtube.com/watch?v=example_NeXTA_demo. This video provides guided instruction on NeXTA's core tools, including network editing workflows, OD filtering techniques, advanced trajectory visualization options, and tensor exploration features.

The demonstration materials are designed to serve multiple audiences: researchers learning to use NeXTA for their own work, practitioners implementing the tool in operational settings, and educators incorporating visualization into their transportation courses.

Impact and Future Directions

NeXTA represents a significant step forward in making sophisticated transportation analysis accessible to a broader community of researchers and practitioners. By combining rigorous mathematical foundations with intuitive visualization interfaces, we've created a tool that bridges the gap between theoretical traffic modeling and practical transportation planning applications.

Looking ahead, we're excited about several emerging directions. The integration of machine learning approaches with traditional traffic flow theory offers promising opportunities for more accurate and robust transportation models (Zhang & Xie, 2011). The growing availability of connected vehicle data will create new possibilities for real-time network analysis and control (Talebpour & Mahmassani, 2016). And the increasing emphasis on multimodal transportation systems will require more sophisticated tools for analyzing the interactions between different travel modes (Cats et al., 2017).

We believe NeXTA's flexible, extensible architecture positions it well to support these evolving needs while maintaining its core strengths in visualization and analysis.

Acknowledgements

The authors thank the anonymous reviewers for their thoughtful feedback, which significantly improved this paper. We're also grateful to the broader GMNS community for their ongoing support and valuable input throughout NeXTA's development. Special thanks to the Federal Highway Administration for making the NGSIM dataset publicly available, enabling much of the validation work presented here.

Beckmann, M., McGuire, C. B., & Winsten, C. B. (1956). *Studies in the economics of transportation*.

Behrisch, M., Bieker, L., Erdmann, J., & Krajzewicz, D. (2011). SUMO – simulation of urban MObility: An overview. *Proceedings of SIMUL 2011, the Third International Conference on Advances in System Simulation*, 63–68. <https://elib.dlr.de/71460/>

Ben-Akiva, M. E., & Lerman, S. R. (1985). *Discrete choice analysis: Theory and application to travel demand*. MIT Press.

Cats, O., Koutsopoulos, H. N., Burghout, W., & Toledo, T. (2017). Dynamic modeling and control of multimodal urban transportation networks. *Transportation Research Part C: Emerging Technologies*, 81, 234–247.

Federal Highway Administration. (2007). *Next generation simulation (NGSIM) vehicle trajectories and supporting data* (FHWA-HRT-07-030). U.S. Department of Transportation.

Federal Highway Administration. (2011). *Trajectory processor user's guide*. U.S. Department of Transportation.

Gazis, D. C. (2002). *Traffic theory*. Springer.

GMNS Development Team. (2020). *General modeling network specification*. <https://github.com/zephyr-data-specs/GMNS>. <https://github.com/zephyr-data-specs/GMNS>

Kim, T., Sharda, S., Zhou, X., & Pendyala, R. M. (2020). A stepwise interpretable machine learning framework using linear regression (LR) and long short-term memory (LSTM): City-wide demand-side prediction of yellow taxi and for-hire vehicle (FHV) service. *Transportation Research Part C: Emerging Technologies*, 120, 102786. <https://doi.org/10.1016/j.trc.2020.102786>

Lighthill, M. J., & Whitham, G. B. (1955). On kinematic waves II. A theory of traffic flow on long crowded roads. *Proceedings of the Royal Society of London. Series A*, 229(1178), 317–345.

McFadden, D. (1974). The measurement of urban travel demand. *Journal of Public Economics*, 3(4), 303–328.

Nevers, B. L., Nguyen, K. M., Quayle, S., Zhou, X., Taylor, J., & others. (2013). *The effective integration of analysis, modeling, and simulation tools*. U.S. Federal Highway Administration, Office of Operations Research; Development.

- 231 Papageorgiou, M. (1991). Dynamic modeling, assignment, and route guidance in traffic
232 networks. *Transportation Research Part B: Methodological*, 25(2–3), 143–159.
- 233 Papageorgiou, M., Diakaki, C., Dinopoulou, V., Kotsialos, A., & Wang, Y. (2003). Review of
234 road traffic control strategies. *Proceedings of the IEEE*, 91(12), 2043–2067.
- 235 Richards, P. I. (1956). Shock waves on the highway. *Operations Research*, 4(1), 42–51.
- 236 Shladover, S., Su, D., & Lu, X.-Y. (2012). Impacts of cooperative adaptive cruise control on
237 freeway traffic flow. *Transportation Research Record*, 2324(1), 63–70.
- 238 Talebpour, A., & Mahmassani, H. S. (2016). Influence of connected and autonomous vehicles
239 on traffic flow stability and throughput. *Transportation Research Part C: Emerging*
240 *Technologies*, 71, 143–163.
- 241 Treiber, M., & Kesting, A. (2013). *Traffic flow dynamics: Data, models and simulation*.
242 Springer.
- 243 Wardrop, J. G., & Whitehead, J. I. (1952). Correspondence. Some theoretical aspects of road
244 traffic research. *Proceedings of the Institution of Civil Engineers*, 1(5), 767–768.
- 245 Yelchuru, B., Kamalanathsharma, R. K., Abdelghany, K., Mahmassani, H. S., Rinelli, P., Li,
246 P., Zhou, X., Ong, B. T., & others. (2017). *Analysis, modeling, and simulation (AMS)*
247 *testbed development and evaluation to support dynamic mobility applications (DMA) and*
248 *active transportation and demand management (ATDM) programs*. U.S. Department of
249 Transportation, Intelligent Transportation Systems Joint Program Office.
- 250 Zhang, Y., & Xie, Y. (2011). A comparative study of three multivariate short-term freeway
251 traffic flow forecasting methods with missing data. *Journal of Intelligent Transportation*
252 *Systems*, 15(4), 205–218.
- 253 Zhou, X., Taylor, J., & Koutsopoulos, H. N. (2012). DTALite: A queue-based mesoscopic
254 traffic simulator for fast model evaluation and calibration. *Proceedings of the 92nd Annual*
255 *Meeting of the Transportation Research Board*.